Repository Appendix B: Detailed audit methods and secondary diagnostics

Supplement to Benchmark-Gated Composite Mean Estimation

Alvaro Rivera-Eraso et al.

Repository Appendix B: Detailed audit methods and secondary diagnostics

B.1 Detailed confirmation and abstention audit design

Fixed-weight confirmation was examined through four complementary audits addressing robustness, bootstrap validity, multiplicity, and possible optimism from candidate selection. These audits did not modify the benchmark gate or the evidence taxonomy. They provided additional checks on candidates that had already completed fixed-weight validation.

The first audit examined the robustness properties implied by the frozen weight vectors. For each confirmed specialist, it recorded the weights assigned to the sample mean, median, and modal components and interpreted them through the influence-function decomposition introduced in Section 2 of the main text. A positive weight on the sample mean implies a formal breakdown point of zero, even when most remaining weight is assigned to estimators with bounded influence. Gross-error sensitivity was assessed relative to the sample mean by examining how the linear term associated with w_{mean} scales the effect of an arbitrarily large observation. Geometric and harmonic components were considered separately because their contributions grow more slowly than the linear contribution of the sample mean.

The second audit addressed bootstrap validity for nonsmooth components. The ordinary bootstrap is unreliable for the mode and may converge slowly for the median [ghosh1984median,romano1988mode]. The practical relevance of this limitation was assessed by inspecting the total weight assigned to modal and median components in each confirmed or otherwise important candidate. The modal weights of the confirmed specialists were very small, whereas candidates with more substantial median weight were retained with an explicit interpretive caveat. An m out of n subsampling interval was identified as an appropriate extension for future confirmation studies.

The third audit examined multiplicity across confirmation candidates. The Benjamini–Hochberg procedure was applied to the candidate-level confirmation signals at a 5% false discovery rate. This correction was an additional audit and did not replace the original dual benchmark gate.

The fourth audit evaluated possible optimism from selecting candidates before reporting their gains. The eight validation seeds were divided into two groups: one half was used for selection and the other for performance reporting. These split-sample estimates were compared with the pooled estimates from all eight seeds. Because each half contained only four seeds, the results were treated as robustness estimates, not as separate inferential intervals.

A complementary audit examined benchmark-retained outcomes. Its purpose was to distinguish genuine abstention from a failure of the genetic algorithm to locate an existing gate-passing region. All benchmark-retained regimes from the expanded discovery stage were evaluated through an independent random Dirichlet search over the same 26-component simplex. The audit covered 17 regimes and both validation modes, producing 34 regime and mode cells.

For each regime and validation mode, the audit generated 4,000 random weight vectors from a symmetric Dirichlet distribution with concentration parameter $\alpha = 0.30$. Every draw was evaluated with the same simulator, admissibility rules, benchmark registry, definition of $q_{0.95}$, and Monte Carlo budget used in fixed-weight validation, with $R = 500$. The original dual gate required improvement in both MSE and $q_{0.95}$, together with stability across all eight validation seeds,

$$303,\ 404,\ 505,\ 606,\ 707,\ 808,\ 909,\ 1010.$$

A regime and mode cell confirmed abstention when no random draw cleared the dual gate at all eight seeds. The frozen weights of the confirmed transfer specialists were included as positive controls to verify that the audit could recover weight vectors already known to satisfy the gate. Randomly identified passing vectors were not treated as confirmed discoveries. They were recorded only as targets for a future search in which their weights would need to be persisted, frozen, and subjected to the complete validation procedure.

B.2 Seed-specific fixed-weight confirmation results

Table 1: Seed-specific fixed-weight confirmation results for the three Lognormal candidates. Mean and $q_{0.95}$ gains are relative to the criterion-specific strongest admissible benchmarks within each validation seed. The $q_{0.95}$ interval is the 95% paired bootstrap interval based on $B = 500$ resamples for that seed. A dual-gate pass requires positive gains in both MSE and $q_{0.95}$.

Candidate	**Mode**	**Seed**	**Mean gain**	$q_{0.95}$ **gain**	$q_{0.95}$ **paired bootstrap 95% CI**	**Dual gate**
CV-019	Locked unseen	303	5.18%	18.36%	[16.36%, 20.12%]	Pass
CV-019	Locked unseen	404	6.15%	17.35%	[12.35%, 23.20%]	Pass
CV-019	Locked unseen	505	7.90%	13.16%	[7.62%, 18.55%]	Pass
CV-019	Locked unseen	606	4.20%	13.73%	[10.40%, 16.79%]	Pass
CV-019	Locked unseen	707	6.88%	18.64%	[16.11%, 19.72%]	Pass
CV-019	Locked unseen	808	6.13%	18.48%	[16.63%, 19.74%]	Pass
CV-019	Locked unseen	909	7.29%	16.78%	[8.37%, 20.98%]	Pass
CV-019	Locked unseen	1010	6.37%	13.96%	[8.97%, 18.79%]	Pass
CV-010	Locked unseen	303	1.93%	2.21%	[1.95%, 2.72%]	Pass
CV-010	Locked unseen	404	2.08%	2.02%	[1.73%, 2.44%]	Pass
CV-010	Locked unseen	505	1.63%	2.13%	[1.57%, 2.65%]	Pass
CV-010	Locked unseen	606	2.23%	2.37%	[1.95%, 2.73%]	Pass
CV-010	Locked unseen	707	2.03%	2.11%	[1.80%, 2.68%]	Pass
CV-010	Locked unseen	808	2.17%	2.30%	[1.96%, 2.80%]	Pass
CV-010	Locked unseen	909	1.67%	2.53%	[2.05%, 3.11%]	Pass
CV-010	Locked unseen	1010	1.99%	2.27%	[1.78%, 2.62%]	Pass
CV-011	Original	303	0.52%	3.41%	[2.75%, 4.05%]	Pass
CV-011	Original	404	-0.45%	3.67%	[2.89%, 4.37%]	Retain
CV-011	Original	505	0.92%	4.45%	[3.62%, 5.04%]	Pass
CV-011	Original	606	-0.06%	3.85%	[3.26%, 4.38%]	Retain
CV-011	Original	707	-0.25%	3.57%	[2.90%, 3.92%]	Retain
CV-011	Original	808	-0.20%	3.91%	[3.34%, 4.62%]	Retain
CV-011	Original	909	-0.65%	3.72%	[2.98%, 4.40%]	Retain
CV-011	Original	1010	-0.55%	3.98%	[3.42%, 4.55%]	Retain

B.3 Seed-level confirmation matrix

B.4 Expanded-basis evidence taxonomy by family

B.5 Candidate-level taxonomy and gain matrix

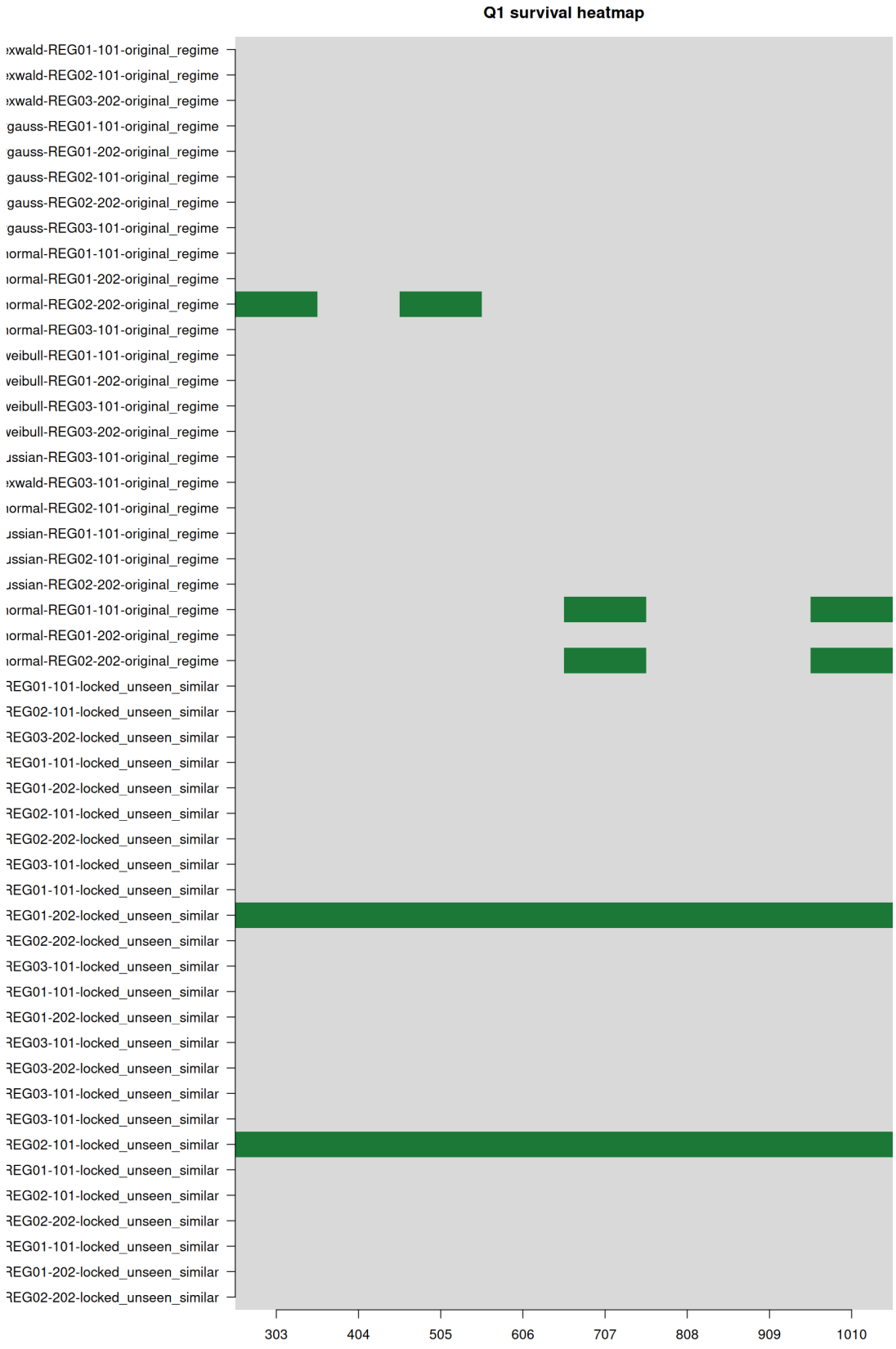


Figure 1: Seed-level expanded-gate survival for the confirmed and local Lognormal signals. Each cell indicates whether a candidate satisfied the expanded benchmark gate for the corresponding validation seed.

Table 2: Candidate-level evidence and gain matrix for the complete 25-candidate expanded-basis validation panel. Positive values indicate improvement and negative values indicate loss relative to the criterion-specific strongest admissible benchmark. Each gain cell reports MSE gain followed by $q_{0.95}$ gain. Discovery values are the gains that defined candidate status during expanded search. Locked-unseen and original values are frozen-weight averages across the eight validation seeds. The CI columns report the number of seeds whose $q_{0.95}$ paired bootstrap interval was entirely positive. "Transfer" denotes transfer specialist; "Discovery-local" denotes an accepted discovery winner not confirmed by the frozen gate; the remaining grades identify diagnostic near-gate and negative-control cells.

Candidate	Family / regime / seed	Source	Taxonomy	Discovery mean / $q_{0.95}$	Locked mean / $q_{0.95}$	Locked $q_{0.95}$ CI seeds	Original mean / $q_{0.95}$	Original $q_{0.95}$ CI seeds
Q1R011 / FWVR011	Weibull / REG01 / 101	GA win	Transfer	+8.30% / +7.32%	+2.79% / +2.72%	8/8	-15.77% / -45.06%	0/8
Q1R012 / FWVR012	Weibull / REG02 / 202	GA win	Transfer	+29.06% / +23.61%	+3.18% / +1.25%	8/8	-15.20% / -34.56%	0/8
FWVR001	Exwald / REG01 / 101	GA win	Discovery-local	+4.69% / +10.44%	-11.06% / -7.36%	0/8	-21.89% / -48.44%	0/8
FWVR002	Exwald / REG03 / 101	GA win	Discovery-local	+23.73% / +11.49%	-32.20% / -58.58%	0/8	-30.24% / -57.94%	0/8
FWVR003	Exwald / REG03 / 202	GA win	Discovery-local	+24.78% / +20.62%	-3.29% / -8.37%	0/8	-18.03% / -40.42%	0/8
FWVR004	Invgauss / REG01 / 101	GA win	Discovery-local	+52.21% / +38.20%	-8.43% / +1.46%	0/8	-36.22% / -90.21%	0/8
FWVR005	Invgauss / REG02 / 101	GA win	Discovery-local	+32.15% / +16.94%	-3.18% / +2.62%	0/8	-57.22% / -164.93%	0/8
FWVR006	Invgauss / REG02 / 202	GA win	Discovery-local	+17.53% / +5.32%	-5.52% / +0.47%	0/8	-42.91% / -103.40%	0/8
FWVR007	Invgauss / REG03 / 202	GA win	Discovery-local	+17.59% / +8.34%	-11.52% / -1.58%	0/8	-24.51% / -50.99%	0/8
FWVR008	Lognormal / REG01 / 101	GA win	Discovery-local	+16.76% / +3.86%	-38.07% / -127.60%	0/8	-24.76% / -85.25%	0/8
FWVR009	Lognormal / REG03 / 101	GA win	Discovery-local	+13.65% / +3.82%	-1.91% / -0.72%	0/8	-43.30% / -164.67%	0/8
FWVR010	Normal / REG02 / 202	GA win	Discovery-local	+1.68% / +6.99%	-0.09% / -0.28%	0/8	-0.03% / -0.54%	0/8
FWVR013	Exgaussian / REG01 / 101	Near-gate	Near-gate	-16.20% / -4.87%	-1.20% / -1.26%	0/8	-1.02% / -1.06%	0/8
FWVR014	Exgaussian / REG02 / 101	Near-gate	Near-gate	-9.84% / -13.01%	-0.95% / -0.98%	0/8	-0.88% / -1.07%	0/8
FWVR015	Exgaussian / REG03 / 101	Near-gate	Near-gate	-10.02% / -8.77%	-1.16% / -1.89%	0/8	-0.84% / -1.10%	0/8
FWVR016	Normal / REG01 / 101	Near-gate	Near-gate	-4.52% / -4.05%	-0.23% / -0.30%	0/8	-0.16% / -0.30%	0/8
FWVR017	Normal / REG02 / 101	Near-gate	Near-gate	-1.57% / -4.51%	-0.15% / -0.42%	0/8	-0.20% / -0.51%	0/8
FWVR018	Weibull / REG02 / 101	Near-gate	Near-gate	-0.64% / -0.75%	-75.95% / -283.58%	0/8	-52.45% / -197.54%	0/8
FWVR019	Weibull / REG03 / 101	Near-gate	Near-gate	-71.64% / -9.71%	+0.66% / -0.15%	2/8	-31.66% / -106.49%	0/8
FWVR020	Exgaussian / REG01 / 202	Control	Negative control	-0.39% / +0.40%	-2.26% / -4.56%	0/8	-2.17% / -4.55%	0/8
FWVR021	Exgaussian / REG02 / 202	Control	Negative control	-1.25% / -2.10%	-0.68% / -0.50%	0/8	-0.95% / -1.05%	0/8
FWVR022	Exgaussian / REG03 / 202	Control	Negative control	-8.44% / -0.66%	-0.44% / -0.60%	0/8	-0.69% / -0.71%	0/8
FWVR023	Normal / REG01 / 202	Control	Negative control	+0.69% / -0.50%	-0.07% / -0.16%	0/8	-0.01% / +0.07%	0/8
FWVR024	Normal / REG03 / 101	Control	Negative control	-1.58% / -2.46%	-0.19% / -0.50%	0/8	-0.19% / -0.24%	0/8
FWVR025	Normal / REG03 / 202	Control	Negative control	-0.13% / -0.21%	-0.12% / -0.25%	0/8	-0.04% / -0.01%	0/8

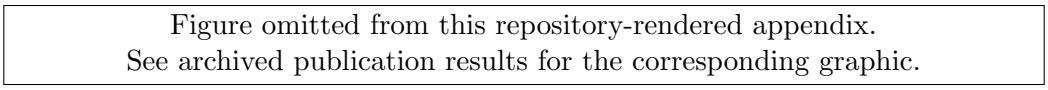


Figure 2: Evidence taxonomy from post-discovery fixed-weight validation ($N_{\mathrm{EST}} = 26$), shown by family. Transfer specialists have interval-supported advantages in locked-unseen evaluation. Discovery-supported local candidates retain valid discovery evidence, near-gate candidates identify borderline regions for diagnosis, and negative-control successes verify appropriate benchmark retention.

B.6 Frozen-weight robustness audit

Table 3: Frozen-weight robustness audit for the two confirmed Weibull transfer specialists. Both estimators substantially attenuate gross-error sensitivity relative to the sample mean. Their small positive sample mean weights preserve a formal breakdown point of zero.

Specialist	w_{mean}	w_{median}	w_{mode}	**Breakdown / gross-error influence**
Weibull REG01 (seed 101)	0.016	0.179	0.007	0 / $\approx 1/62$ of $\bar{X}_n$
Weibull REG02 (seed 202)	0.022	0.018	0.004	0 / $\approx 1/46$ of $\bar{X}_n$

B.7 Random-Dirichlet abstention audit

Table 4: Results of the random Dirichlet abstention audit over the 17 benchmark-retained regimes. The audit comprised 4,000 draws per regime and validation mode, with $\alpha = 0.30$ and stability across eight seeds. The confirmed transfer specialists served as positive controls.

Quantity	**Value**
Abstained regimes audited	17 (34 regime and mode cells)
Regime and mode cells confirming abstention	23/34
Unique regimes fully confirmed in both modes	9/17
Unique regimes with a Dirichlet signal	8/17
Normal / Ex Gaussian controls confirmed	6/6
Positive control, transfer specialists	8/8 and 8/8
Strongest flagged regime, Q1R003 Ex Wald	1.8% pass rate

B.8 External empirical gate funnel

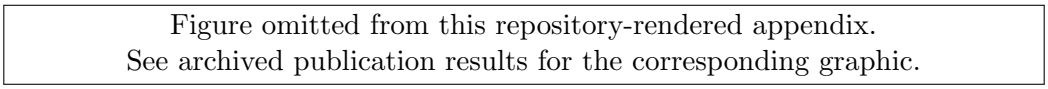


Figure 3: Conjunctive stringency of the external primary gate. The figure annotation reporting 2,211 strong-win cells refers to condition-level rows with positive paired bootstrap intervals among 5,900 audit-ready gate rows. These rows include repeated conditions within the same dataset, specialist, and mode comparison and are not bootstrap replicates or independent evidence. They were aggregated into 809 profile-eligible comparisons; 256 contained at least one strong win supported by a confidence interval, 255 survived Benjamini–Hochberg correction at 5%, and 257 survived at the 10% sensitivity threshold. The 10% set includes one additional strict win and one sensitivity-only row.

Bibliography

Ghosh, Malay; Parr, William C.; Singh, Kesar; Babu, G. Jogesh (1984). A Note on Bootstrapping the Sample Median. The Annals of Statistics, 12, 1130–1135. https://doi.org/10.1214/aos/1176346731.

Romano, Joseph P. (1988). Bootstrapping the Mode. Annals of the Institute of Statistical Mathematics, 40, 565–586. https://doi.org/10.1007/BF00053066.